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Charles R. Doss

Education

University of Washington
Ph.D. in Statistics, Advised by Jon Wellner — September 2008-August 2013

University of Chicago
B.S. in Mathematics (Honors; Phi Beta Kappa, Sigma Xi), — June 2007

Professional Experience

University of Minnesota - Twin Cities, School of Statistics

Director of Graduate Studies	April 2026–Present
Associate Professor	August 2020–Present
Assistant Professor	August 2013–July 2020
Director of Undergraduate Studies	August 2020 – August 2021

Federal Reserve Bank, Chicago

Associate Economist	June 2007-July 2008

Published or Accepted Papers

Doss, C. R. Doubly robust pointwise confidence intervals for a monotonic continuous treatment effect curve. (To appear in *JASA (T&M).*)

Difan Ouyang, Lappui Chung, Mark Osborn, Timothy W. Schacker, and **Doss, C. R.** Dynamic Linear Models for Wastewater-based Epidemiology with Missing Values: an Application to Covid-19 Surveillance. (2025+) To appear in *Data Science in Science.*

Du, J., Chen, L., Xian, X., Luo, A., Tian, F., Wang, G., **Doss, C.R.**, Shen, X., Ding, J. Ice Cream Doesn't Cause Drowning: Benchmarking LLMs Against Statistical Pitfalls in Causal Inference. (2026). *International Conference on Learning Representations (ICLR).*

Doss, C. R. and Huling, J. Doss and Huling's contribution to the discussion of 'Augmented balancing weights as linear regression' by Bruns-Smith, Dukes, Feller, and Ogburn. (2025+), *JRSS-B.*

Doss, C. R., Weng, G., Wang, L., Moscovice, I., and Chantarat, T. A nonparametric doubly robust approach to testing for a continuous treatment effect. (2024) *The Annals of Statistics.* **52**(4), 1592–1615.

Doss, C. R., and McFowland III, E. (2022) Non-parametric subset scanning for detection of heteroscedasticity. *The Journal of Computational and Graphical Statistics.* **31**(3), 813–823.

Balabdaoui, F. **Doss, C. R.**, and Durot, C. (2021) Unlinked monotone regression. *The Journal of Machine Learning Research.* **22**(172).

Molstad, A. J., Weng, G., **Doss, C.R.**, and Rothman, A. J. (2021) An explicit

mean-covariance parameterization for multivariate response linear regression. *The Journal of Computational and Graphical Statistics.* **30**(3), 612–621.

Doss, C. R. (2020) Bracketing numbers of convex and m-monotone functions on polytopes. *The Journal of Approximation Theory.* **256**, 105425.

Doss, C. R. and Wellner J. A. (2019). Univariate log-concave density estimation with symmetry or modal constraints. *The Electronic Journal of Statistics*, **13**(2), 2391–2461.

Doss, C. R. and Wellner J. A. (2019). Inference for the mode of a log-concave density. *The Annals of Statistics*, **47**(5), 2950–2976.

Doss, C. R. (2019). Concave regression: value-constrained estimation and likelihood ratio-based inference. *Mathematical Programming.* **174**(1-2), 5–39.

Doss, C. R. and Weng, G. (2018). Bandwidth selection for kernel density estimators of multivariate level sets and highest density regions. *The Electronic Journal of Statistics*, **12**:4313-4376.

Balabdaoui, F. and **Doss, C. R.** (2018) Inference for a mixture of symmetric distributions under log-concavity. *Bernoulli*, **24**(2):1053–1071.

Doss, C. R. and Wellner J. A. (2016) Global rates of convergence of the MLEs of log-concave and s-concave densities. *The Annals of Statistics*, **44**:954–981.

Doss, C.R., Flegal, J. M., Jones, G.L. and Neath R. C. (2014). Markov chain Monte Carlo estimation of quantiles. *The Electronic Journal of Statistics,* **8**:2448-2478.

Doss, C. R., Suchard, M. A., Holmes, I., Kato-Maeda, M., and Minin, V. N.. (2013). Fitting birth-death processes to panel data with applications to bacterial DNA fingerprinting. *The Annals of Applied Statistics.* **7**:2315-2335.

Interdisciplinary/Application publications:

Doss, C. R., Mark J. Osborn, Stacey Stark, Joshua Rhein, Jacalynn Donkersgoed, Donna Budde, Shannon Champeau, Carolyn Meyer, Mason Hayden, Laura Landini, Difan Ouyang, Lappui Chung, Yi Tang, Sara Vetter, and Timothy W. Schacker. Wastewater Measures of SARS-CoV-2 Accurately Predict Frequency of Symptomatic Infections in the Community. *Journal of Infectious Diseases*, Volume 232, Issue 3, 15 September 2025, Pages e459–e465.

Osborn, Mark J., Shannon Champeau, Carolyn Meyer, Mason Hayden, Laura Landini, Stacey Stark, Stephanie Preekett, Sara Vetter, Zachary Zirnhelt, Stephanie Meyer, Daniel Huff, Timothy W. Schacker, **Doss, C. R.**. (2024) Wastewater Surveillance of SARS-CoV-2 in Minnesota. *Water* **16**(4), 541.

Griffing, D., Larson, S., Hollander, J., Christiansen, J., and **Doss, C. R.** (2014). Observations on abundance of bluntnose sixgill sharks, Hexanchus griseus, in an urban waterway in Puget Sound, 2003-2005. *PloS one* 9.1: e87081.

Submitted Papers

Ham, D.Y., Westling, T., **Doss, C. R.** Doubly robust estimation and inference for a log-concave counterfactual density. (Submitted.)

Doss, C.R and VandenBerg, R.O. Likelihood Ratio Tests for Monotonic Functions with Regions of Flatness. (Submitted.)

Difan Ouyang, Keeley J. Morris, Mark J. Osborn, Timothy W. Schacker, **Doss,**

C. R. Viral Lineage Features Can Enhance the Detection of COVID-19 Surges via Machine Learning. (Submitted.)

Doss, C.R. and Zhu, Q. Active Sampling for change-point detection. (Submitted.)

Preprints (near submission)

Zhang, R., Doss, C.R., and Huling. J. Multiply robust estimation of heterogeneous principal strata effects under principal ignorability.

Doss, C.R. and Ouyang, D. Nonparametric changepoint detection for multivariate distributions using optimal transportation.

Ham, D. and Doss, C.R. Inference for the argmax of a concave function.

Doss, C.R and VandenBerg, R.O. Confidence bands for a continuous treatment effect curve based on shape constraints.

Funding & Awards

- MN Dept. Health, co-PI, July 2023 – July 2024
- NSF DMS-2210312 grant (sole PI) July 2022—June 2025
- University of Minnesota (College of Liberal Arts) Warwick Mid-Career Faculty Research Award Mar 2022–Dec 2023
- Minnesota Dept. of Health grant June 2021–Dec 2021
  Monitoring of COVID-19 through wastewater-based epidemiology. I received funding for $> 25\%$ salary and for 2 RA's for June 2021–present
- NSF DMS-1712664 grant (sole PI) July 2017—June 2020
- University of Minnesota, Grant-in-Aid ($35,702) 2016–17

Invited Talks

JSM, *Nonparametric doubly robust (almost pivotal) pointwise inference for monotonic continuous treatment effects,* August 2025, Nashville, TN, USA.

NSF@75, *Nonparametric doubly robust (almost pivotal) pointwise inference for monotonic continuous treatment effects,* July 2025, Online.

WNAR, *Nonparametric doubly robust (almost pivotal) pointwise inference for monotonic continuous treatment effects,* June 2025, Whistler, Canada.

CMstatistics, *Doubly robust almost pivotal pointwise confidence intervals for a monotonic continuous treatment effect curve*, Dec 2024, London, England.

Joint Statistical Meetings, *Doubly robust almost pivotal pointwise confidence intervals for a monotonic continuous treatment effect curve,* Aug 2024, Portland, Oregon.

International Symposium on Nonparametric Statistics, *Doubly robust estimation and inference for a log-concave counterfactual density,* June 2024, Braga Portugal.

[Cancelled] CMStatistics, *Nonparametric doubly robust confidence intervals for a monotonic continuous treatment effect curve,* December 2023, Berlin, Germany.

Joint Statistical Meetings, *Sequential Monte Carlo for fitting state-space SIR compartment models,* August 2023, Toronto, Canada.

New England Statistics Symposium, *A nonparametric doubly robust test for a continuous treatment effect based on observational data,* June 2023. Boston, Massachusetts, United States.

A nonparametric doubly robust test for a continuous causal treatment effect, 2023, George Mason University, Fairfax, Virginia, United States

A nonparametric doubly robust test for a continuous causal treatment effect, 2023, University of South Carolina, Columbia, South Carolina, United States.

Nonparametric doubly robust testing for continuous treatment effects via smoothness and via shape constraints, August 2022, JSM, Washington D.C.

(Cancelled due to COVID) *A nonparametric doubly robust test for a continuous treatment effect*, May 2022, Robustness and Resilience in Stochastic Optimization and Statistical Learning, Erice, Italy.

A nonparametric doubly robust test for a continuous treatment effect, April 2022, University of Massachusetts (Amherst) Statistics, Amherst, MA.

A nonparametric doubly robust test for a continuous treatment effect, Mar 2022, Columbia Statistics, New York, NY.

Discussion of "Optimal Subgroup Selection" presented by Richard J. Samworth, Jan 2022, International Seminar on Selective Inference (virtual/online).

A nonparametric doubly robust test for a continuous treatment effect, December 2021, ERCIM/CMStatistics, London, United Kingdom.

Discussion of "Recent advances in unlinked and permuted regression", August 2021, JSM (virtual).

Unlinked monotone regression, October 2020, George Mason University Statistics Seminar, (virtual) Fairfax, VA.

In 2020 two invited talks were canceled due to COVID-19 and travel difficulties.

Unlinked monotone regression, Aug 2020, JSM 2020, Philadelphia, PA. (Virtual due to COVID-19).

Likelihood ratio tests and confidence intervals based on the shape constraint of concavity, Dec 2019, ERCIM/CMStatistics, London, United Kingdom.

Likelihood ratio tests and confidence intervals based on the shape constraint of concavity, Nov 2019, Duke University Statistical Science Seminar, Durham, NC.

Likelihood ratio tests and confidence intervals based on the shape constraint of concavity, May 2018, 2018 IISA International Conference on Statistics, Gainesville, FL.

Bandwidth selection for kernel density estimators of multivariate level sets and highest density regions, Feb 2018, Data Science Seminar, Institute for Mathematics and its Applications, University of Minnesota.

Bandwidth selection for kernel density estimators of multivariate level sets and highest density regions, Jan 2018, Shape-Constrained Methods: Inference, Applications, and Practice. Banff, Canada.

Estimation and Computation of Log-Concave Densities, May 2017, SIAM Conference on Optimization. Vancouver, Canada.

Inference for a Mixture of Symmetric Distributions under Log-Concavity, March 2017, University of Minnesota Statistics Department Seminar. Minneapolis, MN.

Inference for a Mixture of Symmetric Distributions under Log-Concavity, June

2016, International Workshop on Applied Probability. Toronto, Canada.

Inference for a Mixture of Symmetric Distributions under Log-Concavity, March 2016, Colorado State University Statistics Seminar. Fort Collins, CO.

Nonparametric Inference about a Density's Mode via the Log-Concave Shape Constraint, January 2016, University of Minnesota Student Seminar. Minneapolis, MN.

Shape-constrained Inference for the Argmax of a Concave Function, December 2015, ERCIM/CMStatistics. London, United Kingdom.

A Likelihood Ratio Test for the Mode of a Log-Concave Density, November 2015, University of Michigan Seminar. Ann Arbor, MI.

A Likelihood Ratio Test for the Location of the Maximum of a Concave Function, October 2015, Shape Constrained Inference: Open Problems and New Directions, Leiden, Netherlands.

Symmetric Log-Concave Density Estimation and Mixture Modeling, August 2015, Joint Statistical Meetings, Seattle, Washington.

A Likelihood Ratio Test for the Location of the Maximum of a Concave Function, July 2015, European Meeting of Statisticians, Amsterdam, Netherlands.

Nonparametric Inference about a Density's Mode and Shape Constraints, August 2014, Joint Statistical Meetings, Boston, USA.

Log-Concavity and Nonparametric Inference about a Density's Mode, August 2014, IMS New Researcher's Conference, Boston, USA.

Semiparametric Modeling with Log-Concave Densities, May 2014, Statistical Society of Canada Annual Meeting, Toronto, Canada.

Nonparametric Mode Estimation via the Log-Concave Shape Constraint, October 2011, U. Heidelberg Applied Math Dept., Heidelberg, Germany.

Courses Taught

Theory of Statistics I (STAT 4101)	Fall 2013–15, 2017
Theory of Statistics II (STAT 4102)	Spring 2014–17, Fall 2018–19
Ph.D. Asymptotics (STAT 8112)	Spring 2016–19
Time Series Analysis (STAT 5511)	Spring 2018–21, Fall 2021–22
Ph.D. Literature Seminar (STAT 8913)	Fall 2019
Theory of Statistics I (STAT 5101)	Spring 2021, Fall 2021
Topics Course (STAT 8931)	Spring 2023

Supervising

Current Ph.D. students I supervise:

- Zexian Lu
- Robert Oliver VandenBerg
- Alexander Kuhn

Ph.D. Theses Advised/Completed:

- Guangwei Weng

Ph.D. Thesis Committee Member:

- Dae Young Ham
- Martin Garcia Vasquez (Economics Ph.D.)
- Yang Song
- Evan Olawsky
- Sanhita Sengupta
- Wenjun Lang
- Jiawei Zhang
- Ning Dai
- Ming Gao
- June Park (Mathematics)
- Daniel Eck
- Patrick Schnell (Biostatistics)
- Megan Heyman
- Felipe Acosta Archila

Master's thesis ("Plan B") Supervisor:

- King Yiu Suen
- Hexuan Zhang (Data Science; 2021–2022)
- Difan Ouyang (2020–2022)
- Zachary Gunderson (2018)
- Jiuwu Jiu (2016)
- Qimeng Chen (2015)

Master's thesis ("Plan B") Committee Member:

- Yi Rong
- Patrick Skoglund
- Rihan Chen (2016)
- Jingxiang Li (2016)
- Jing Yang (2015)

Departmental Service

Seminar Coordinator, U. Minnesota — Fall 2014

PhD Exam Committee, U. Minnesota — 2013–16, 17–19

- Chair — 2015–16, 17–18

Fellowships Committee — 2016-17

Admissions Committee — 2017–18

Search/Hiring Committee — 2018–20, 22–23

- Chair — 2019–20

Student Recruiting, Retention, and Support (Diversity, Equity, and Inclusion) Committee — 2020–21

- Chair — 2020–21

Curriculum Committee, Chair 2022–23

Director of Undergraduate Studies Aug 2020–Aug 2021

Subcommittee Member

- MS Curriculum subcommittee, U. Minnesota Fall 2013
- Undergraduate Curriculum subcommittee, U. Minnesota Fall 2014
- Subcommittee on SLOs, U. Minnesota Spring 2016
- Curriculum subcommittee (PhD Exams), U. Minnesota Spring 2017
- Reforming Ph.D. exam subcommittee, U. Minnesota Fall 2018
- Three year plan subcommittee for undergraduate education, U. Minnesota Fall 2021

Webmaster, U. Washington September 2010–March 2012

College & University Service

U. Minn. CLA Faculty Research Awards Committee 2016–19

U. Minn. CLA General Assembly 2017-19

Professional Service

Associate Editor, The Electronic Journal of Statistics 2022–present

Associate Editor, The American Statistician 2020–present

IMS Committee 2024

NSF DMS Statistics panel reviewer 2023, 2024

Mentor with Math Alliance 2020–Present

Scientific Program Committee, CMStatistics conference 2023

Organizer, UMN Statistics presence at recruiting events 2019–20
Organized UMN presence and attended (virtual, due to COVID) recruiting several national events aimed at increasing the presence of different underrepresented students in UMN Statistics

Organizer, Machine Learning Summer School Day Camp 2019–
All-girls week-long session and all-gender week-long session

Ad-hoc Committee forming the IMS New Researchers Group 2014–15

Committee member, IMS New Researchers Group 2015–16

Organizer, "Statistics day" (for local high schoolers) 2009

Served as Referee for:
The American Statistician
Annals of Statistics
Annals of Applied Statistics
Journal of the American Statistical Association
Journal of the Royal Statistical Society (Series B)
Bernoulli
Bulletin of the Malaysian Mathematical Sciences Society
Communications in Statistics
Conference on Learning Theory (COLT)
Electronic Journal of Statistics
IEEE Conference on Data Science

Information and Inference
Journal of Nonparametric Statistics
Mathematical Programming,
Sankhya, the Indian Journal of Statistics (Series A)
Scandinavian Journal of Statistics
SIAM Journal on Optimization
Statistics and Data Mining
Statistical Science
Statistica Sinica
Journal of Statistical Planning and Inference
Statistics and Probability Letters.

Served as ad-hoc reviewer for NSF MMS submission

Software

DOBAD R Package

logcondens.mode R Package

lsbs R package

UMR R package

DRDRtest R package

HSS R package

DCDesign R package (`https://github.com/taalbrecht/DCDesign`)